COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	9/2/2026	**DATE SUBMITTED:**	9/2/2026
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ACTIVITY NO. ___ PRELIM SKILLS EXAM

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Problem: Daily Step Goal Tracker

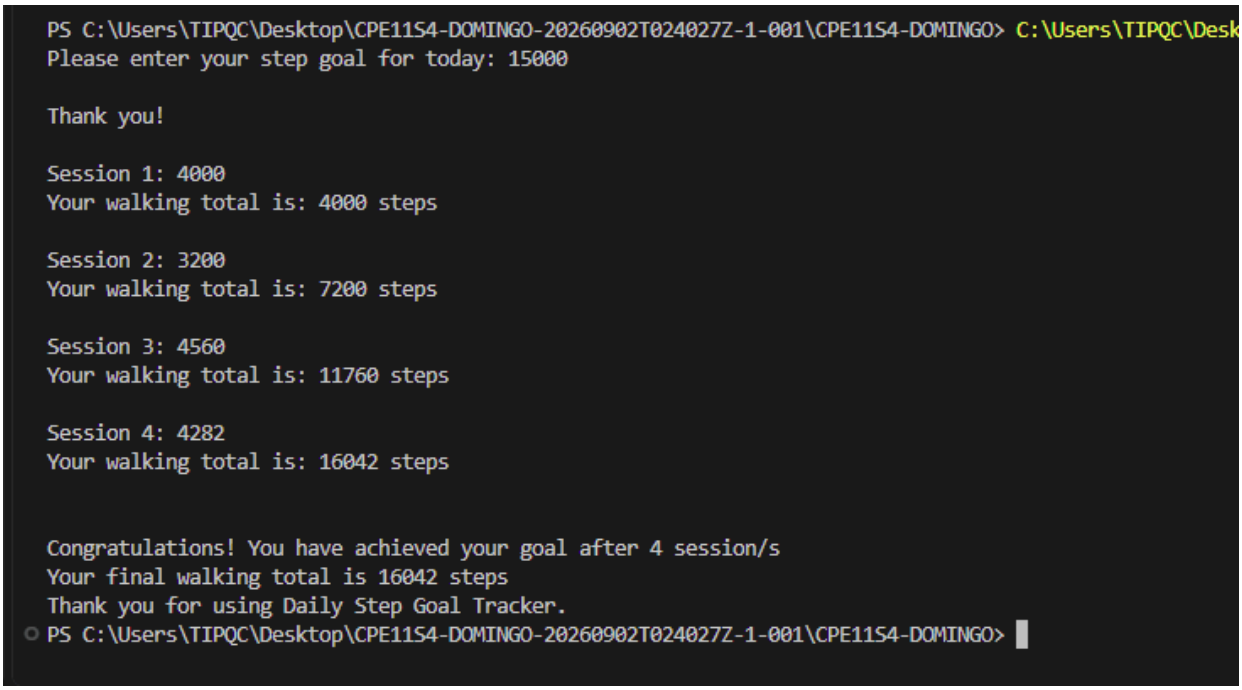
You are tasked with developing a Daily Step Goal Tracker for a fitness app used by casual walkers. Users log their steps in batches throughout the day (e.g., after each walk) instead of all at once, and the app needs to tell them when they've hit their goal. To make their job easier, create a program that:

- asks the user to enter their daily step goal before logging begins.
- repeatedly asks the user to enter the number of steps taken in a session.
- keeps a running total of steps logged so far, adding each new session's steps to it.
- After each session is entered, displays the current running total.
- Continues asking for session steps, using a loop, until the running total meets or exceeds the daily goal.
- Once the goal is reached or exceeded, displays a congratulations message along with how many sessions it took and the final total step count.

CODE:

```
12  #include <iostream>
13  using namespace std;
14
15  int main () {
16
17      int num1, num2, total = 0, g = 0;
18
19      cout << "Welcome Walker!" << endl;
20      cout << "Please enter your step goal for today: ";
21      cin >> num1;
22
23      cout << endl;
24      cout << "Thank you!" << endl;
25      cout << endl;
26
27      while (total < num1) {
28
29          g++;
30          cout << "Session " << g << ": ";
31          cin >> num2;
32
33          total += num2;
34          cout << "Your walking total is: " << total << " steps" << endl;
35          cout << endl;
36
37      }
38      cout << endl;
39      cout << "Congratulations! You have achieved your goal after " << g << " session/s" << endl;
40      cout << "Your final walking total is " << total << " steps" << endl;
41      cout << "Thank you for using Daily Step Goal Tracker.";
42
43
44      return 0;
45  }
```

SAMPLE OUTPUT:



EXPLANATION: Problem 1 asked for a program that calculates the running total for each session of a person walking until the typed goal by the person is either reached or surpassed. The code starts with asking the user to input the step goal which will then proceed to a while loop that asks the user how many steps he/she took after a session, it will all then be added up until the condition is satisfied (running total is greater than or equal than the step goal. If the condition is satisfied, the program will display the final walking total along with the number of sessions it took.

2. **Problem: Workout Duration Validator**

You are tasked with adding an input-validation step to the same fitness app. Before a workout can be logged, the app needs to make sure the user enters a sensible duration — this exercises a *validate-and-reject* loop rather than a running-total loop. Create a program that:

- repeatedly asks the user to enter their planned workout duration in minutes.
- checks whether the entered value is valid, it must be greater than 0 and no more than 180 minutes.
- if the value is invalid, displays an error message explaining the valid range, and asks again.
- Continues asking, using a loop, until the user enters a valid duration.
- Once a valid duration is entered, displays a confirmation message showing the workout duration and how many attempts it took to enter a valid value.

CODE:

```cpp
#include <iostream>
using namespace std;

int main () {

    int num1, g = 1;

    cout << "Welcome to Workout Duration Validator!" << endl;
    cout << "Please type in your workout duration in minutes." << endl;
    cout << endl;

    cout << "Attempt " << g << ": ";
    cin >> num1;

    while (num1 > 180 || num1 <= 0) {

        g++;
        cout << "Invalid Amount! Please try again (Allowed value: 1 to 180 minutes)." << endl;
        cout << endl;
        cout << "Attempt " << g << ": ";
        cin >> num1;

    }
    cout << endl;
    cout << "Your workout duration is valid!" << endl;
    cout << "Workout duration: " << num1 << " minutes"<< endl;
    cout << "Number of attempts until a valid value: " << g << " attempt/s" << endl;
    cout << "Thank you for using Workout Duration Validator!";




    return 0;
}
```

SAMPLE OUTPUT:

```
Welcome to Workout Duration Validator!
Please type in your workout duration in minutes.

Attempt 1: 535
Invalid Amount! Please try again (Allowed value: 1 to 180 minutes).

Attempt 2: 3242
Invalid Amount! Please try again (Allowed value: 1 to 180 minutes).

Attempt 3: 3525
Invalid Amount! Please try again (Allowed value: 1 to 180 minutes).

Attempt 4: 0
Invalid Amount! Please try again (Allowed value: 1 to 180 minutes).

Attempt 5: -123
Invalid Amount! Please try again (Allowed value: 1 to 180 minutes).

Attempt 6: 121

Your workout duration is valid!
Workout duration: 121 minutes
Number of attempts until a valid value: 6 attempt/s
Thank you for using Workout Duration Validator!
PS C:\Users\TIPQC\Desktop\CPE11S4-DOMINGO-20260902T024027Z-1-001\CPE11S4-DOMINGO>
```

EXPLANATION: In this problem, the user is trying to determine the validity of his/her workout duration. Hence, this program asks the user for his/her own tentative time duration of workout and determines whether it is valid or not. The parameters that indicate that the duration is valid is when the duration in minutes is greater than 0 but no more than 180 minutes. If the value that the user typed is invalid, the program will repeat on asking the user to try again until there is an appropriate value that satisfies the condition. A counter is also placed in the code to count the number of attempts taken before obtaining a valid workout duration.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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